CORE PURE / FURTHER CALCULUS

Improper Integrals and Mean Values

Worked solutions 16 questions · 116 marks

Questions

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1. The function f is defined by

$$\mathrm{f}(x) = \frac{x}{x^2+1}, \qquad 1 \leq x \leq 3$$

Find the mean value of f over the interval $1 \leq x \leq 3$.

Give your answer in the form $p \ln q$, where p is a rational number and q is an integer. **[4]**

Solution

The mean value of $\mathrm{f}(x)$ over $a \leq x \leq b$ is

$$\frac{1}{b-a}\int_a^b \mathrm{f}(x)\,\mathrm{d}x$$

So here the mean value is

$$\frac{1}{3-1}\int_1^3 \frac{x}{x^2+1}\,\mathrm{d}x$$

The numerator is a multiple of the derivative of the denominator, since $\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right) = 2x$. Also $x^2+1 > 0$ for all x, so the logarithm needs no modulus signs. Hence

$$\begin{aligned}\int_1^3 \frac{x}{x^2+1}\,\mathrm{d}x &= \frac{1}{2}\int_1^3 \frac{2x}{x^2+1}\,\mathrm{d}x \\ &= \left[\frac{1}{2}\ln\left(x^2+1\right)\right]_1^3 \\ &= \frac{1}{2}\ln 10 - \frac{1}{2}\ln 2 \\ &= \frac{1}{2}\ln 5\end{aligned}$$

Therefore the mean value is

$$\frac{1}{2} \times \frac{1}{2}\ln 5 = \frac{1}{4}\ln 5$$

This is in the form $p \ln q$ with $p = \frac{1}{4}$ and $q = 5$.

2. Show that

$$\int_2^\infty \frac{x}{(x^2+5)^{\frac{3}{2}}}\,\mathrm{d}x = \frac{1}{3}$$

You must show the limiting process clearly. **[5]**

Solution

Replace the infinite upper limit by t and take the limit:

$$\int_2^\infty \frac{x}{(x^2+5)^{\frac{3}{2}}}\,\mathrm{d}x = \lim_{t\to\infty}\int_2^t x\left(x^2+5\right)^{-\frac{3}{2}}\,\mathrm{d}x$$

The integrand is a multiple of $f'(x)(f(x))^{-\frac{3}{2}}$ with $f(x)=x^2+5$, so use the reverse chain rule. Differentiating gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+5\right)^{-\frac{1}{2}} = -\frac{1}{2}\times 2x\left(x^2+5\right)^{-\frac{3}{2}} = -x\left(x^2+5\right)^{-\frac{3}{2}}$$

so

$$\int x\left(x^2+5\right)^{-\frac{3}{2}}\,\mathrm{d}x = -\left(x^2+5\right)^{-\frac{1}{2}} + c = -\frac{1}{\sqrt{x^2+5}} + c$$

For $t > 2$:

$$\begin{aligned}\int_2^t \frac{x}{(x^2+5)^{\frac{3}{2}}}\,\mathrm{d}x &= \left[-\frac{1}{\sqrt{x^2+5}}\right]_2^t \\ &= -\frac{1}{\sqrt{t^2+5}} + \frac{1}{\sqrt{4+5}} \\ &= \frac{1}{3} - \frac{1}{\sqrt{t^2+5}}\end{aligned}$$

As $t\to\infty$, $\sqrt{t^2+5}\to\infty$, so $\frac{1}{\sqrt{t^2+5}}\to 0$. Therefore the limit exists and

$$\int_2^\infty \frac{x}{(x^2+5)^{\frac{3}{2}}}\,\mathrm{d}x = \lim_{t\to\infty}\left(\frac{1}{3} - \frac{1}{\sqrt{t^2+5}}\right) = \frac{1}{3} - 0 = \frac{1}{3}$$

as required.

3. By first expressing

$$\frac{4}{x(x+1)(x+2)}$$

in partial fractions, show that the improper integral

$$\int_1^\infty \frac{4}{x(x+1)(x+2)}\,\mathrm{d}x$$

converges, and find its exact value in the form $\ln k$, where k is a rational number to be determined. **[8]**

Solution

Write the integrand in partial fractions:

$$\frac{4}{x(x+1)(x+2)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2}$$

Multiplying through by $x(x+1)(x+2)$ gives

$$4 = A(x+1)(x+2) + Bx(x+2) + Cx(x+1)$$

Substituting values of x:

$$\begin{aligned} x = 0: &\quad 4 = 2A &\implies &\quad A = 2 \\ x = -1: &\quad 4 = -B &\implies &\quad B = -4 \\ x = -2: &\quad 4 = 2C &\implies &\quad C = 2 \end{aligned}$$

So

$$\frac{4}{x(x+1)(x+2)} = \frac{2}{x} - \frac{4}{x+1} + \frac{2}{x+2}$$

The integral has an infinite upper limit, so replace it with t and consider the limit as $t \to \infty$. For $x \geq 1$, the expressions x, $x+1$ and $x+2$ are all positive, so no modulus signs are needed in the logarithms. For $t > 1$:

$$\begin{aligned} \int_1^t \frac{4}{x(x+1)(x+2)}\,\mathrm{d}x &= \int_1^t \left(\frac{2}{x} - \frac{4}{x+1} + \frac{2}{x+2}\right)\mathrm{d}x \\ &= \Big[2\ln x - 4\ln(x+1) + 2\ln(x+2)\Big]_1^t \\ &= \left[2\ln\left(\frac{x(x+2)}{(x+1)^2}\right)\right]_1^t \\ &= 2\ln\left(\frac{t(t+2)}{(t+1)^2}\right) - 2\ln\left(\frac{1\times 3}{2^2}\right) \\ &= 2\ln\left(\frac{t^2+2t}{t^2+2t+1}\right) - 2\ln\left(\frac{3}{4}\right) \end{aligned}$$

Dividing the numerator and denominator by t^2:

$$\frac{t^2+2t}{t^2+2t+1} = \frac{1+\frac{2}{t}}{1+\frac{2}{t}+\frac{1}{t^2}} \to \frac{1+0}{1+0+0} = 1 \quad \text{as } t \to \infty$$

Hence

$$2\ln\left(\frac{t^2+2t}{t^2+2t+1}\right) \to 2\ln 1 = 0 \quad \text{as } t \to \infty$$

The limit exists and is finite, so the improper integral converges. Its value is

$$\begin{aligned}\int_1^\infty \frac{4}{x(x+1)(x+2)}\,\mathrm{d}x &= \lim_{t\to\infty}\left[2\ln\left(\frac{t^2+2t}{t^2+2t+1}\right)-2\ln\left(\frac{3}{4}\right)\right]\\ &= 0-2\ln\left(\frac{3}{4}\right)\\ &= 2\ln\left(\frac{4}{3}\right)\\ &= \ln\left(\frac{16}{9}\right)\end{aligned}$$

This is in the form $\ln k$ with $k=\frac{16}{9}$.

4. Let

$$y = \frac{1}{1 - \sin x}$$

By first writing y in terms of $\sec x$ and $\tan x$, find the exact mean value of y with respect to x over the interval $0 \leq x \leq \frac{1}{6}\pi$. **[6]**

Solution

Multiply the numerator and denominator by $1 + \sin x$:

$$\begin{aligned} y &= \frac{1 + \sin x}{(1 - \sin x)(1 + \sin x)} \\ &= \frac{1 + \sin x}{1 - \sin^2 x} \\ &= \frac{1 + \sin x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} + \frac{1}{\cos x} \times \frac{\sin x}{\cos x} \\ &= \sec^2 x + \sec x \tan x \end{aligned}$$

The mean value of y over $a \leq x \leq b$ is

$$\frac{1}{b - a} \int_a^b y \, \mathrm{d}x$$

Using $\frac{\mathrm{d}}{\mathrm{d}x}(\tan x) = \sec^2 x$ and $\frac{\mathrm{d}}{\mathrm{d}x}(\sec x) = \sec x \tan x$:

$$\begin{aligned} \int_0^{\frac{\pi}{6}} \left(\sec^2 x + \sec x \tan x\right) \mathrm{d}x &= \left[\tan x + \sec x\right]_0^{\frac{\pi}{6}} \\ &= \left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}\right) - (0 + 1) \\ &= \frac{3}{\sqrt{3}} - 1 \\ &= \sqrt{3} - 1 \end{aligned}$$

The interval has length $\frac{\pi}{6}$, so the mean value is

$$\frac{1}{\frac{\pi}{6}}\left(\sqrt{3} - 1\right) = \frac{6\left(\sqrt{3} - 1\right)}{\pi}$$

5. The power drawn, p watts, by an electric motor at time t hours after it is switched on can be modelled by the equation

$$p = 40\tanh^2\left(\tfrac{1}{2}t\right), \qquad t \geq 0$$

Using an appropriate hyperbolic identity, find, according to the model, the **mean** power drawn by the motor over the first 2 hours after it is switched on. Give your answer as an exact value in the form $\frac{a}{e^2+b}$, where a and b are integers. **[6]**

Solution

The mean value of p over $t_1 \leq t \leq t_2$ is

$$\frac{1}{t_2 - t_1}\int_{t_1}^{t_2} p\,\mathrm{d}t$$

So the mean value of p over $0 \leq t \leq 2$ is

$$\frac{1}{2-0}\int_0^2 40\tanh^2\left(\tfrac{1}{2}t\right)\mathrm{d}t$$

Dividing the identity $\cosh^2 x - \sinh^2 x = 1$ by $\cosh^2 x$ gives

$$1 - \tanh^2 x = \operatorname{sech}^2 x \quad \Longrightarrow \quad \tanh^2 x = 1 - \operatorname{sech}^2 x$$

So, with $x = \frac{1}{2}t$, the mean value is

$$20\int_0^2 \left(1 - \operatorname{sech}^2\left(\tfrac{1}{2}t\right)\right)\mathrm{d}t$$

Since $\frac{\mathrm{d}}{\mathrm{d}t}\tanh\left(\frac{1}{2}t\right) = \frac{1}{2}\operatorname{sech}^2\left(\frac{1}{2}t\right)$, we have $\int \operatorname{sech}^2\left(\frac{1}{2}t\right)\mathrm{d}t = 2\tanh\left(\frac{1}{2}t\right) + c$. Hence the mean value is

$$\begin{aligned} 20\left[t - 2\tanh\left(\tfrac{1}{2}t\right)\right]_0^2 &= 20(2 - 2\tanh 1) - 20(0 - 0) \\ &= 40(1 - \tanh 1) \end{aligned}$$

Now write $\tanh 1$ in exponential form:

$$\tanh 1 = \frac{e - e^{-1}}{e + e^{-1}} = \frac{e^2 - 1}{e^2 + 1}$$

so

$$1 - \tanh 1 = \frac{(e^2+1) - (e^2-1)}{e^2+1} = \frac{2}{e^2+1}$$

Therefore the mean value is

$$40 \times \frac{2}{e^2+1} = \frac{80}{e^2+1}$$

The mean power drawn over the first 2 hours is $\frac{80}{e^2+1}$ W, which is in the form $\frac{a}{e^2+b}$ with $a = 80$, $b = 1$.

6. **(a)** Explain why

$$\int_e^\infty \frac{\ln x}{x^3}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) Use integration by parts to find

$$\int \frac{\ln x}{x^3}\,\mathrm{d}x$$ **[3]**

(c) Hence show that

$$\int_e^\infty \frac{\ln x}{x^3}\,\mathrm{d}x$$

converges, and find its exact value. You should show the limiting process used, and state clearly the behaviour of any limit you rely on. **[4]**

Solution

(a) The upper limit of integration is infinite, so the interval of integration is unbounded and the integral is improper.

(b) Integration by parts states

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Take $u = \ln x$, $\frac{\mathrm{d}v}{\mathrm{d}x} = x^{-3}$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$, $v = -\frac{1}{2x^2}$. Then

$$\begin{aligned}\int \frac{\ln x}{x^3}\,\mathrm{d}x &= -\frac{\ln x}{2x^2} - \int\left(-\frac{1}{2x^2}\right)\frac{1}{x}\,\mathrm{d}x\\ &= -\frac{\ln x}{2x^2} + \frac{1}{2}\int x^{-3}\,\mathrm{d}x\\ &= -\frac{\ln x}{2x^2} + \frac{1}{2}\left(-\frac{1}{2x^2}\right) + c\\ &= -\frac{\ln x}{2x^2} - \frac{1}{4x^2} + c\end{aligned}$$

(c) Replace the infinite upper limit by t, where $t > e$, and let $t \to \infty$:

$$\int_e^\infty \frac{\ln x}{x^3}\,\mathrm{d}x = \lim_{t\to\infty}\int_e^t \frac{\ln x}{x^3}\,\mathrm{d}x$$

Using part (b), and $\ln e = 1$:

$$\begin{aligned}\int_e^t \frac{\ln x}{x^3}\,\mathrm{d}x &= \left[-\frac{\ln x}{2x^2} - \frac{1}{4x^2}\right]_e^t\\ &= \left(-\frac{\ln t}{2t^2} - \frac{1}{4t^2}\right) - \left(-\frac{1}{2e^2} - \frac{1}{4e^2}\right)\\ &= \frac{3}{4e^2} - \frac{\ln t}{2t^2} - \frac{1}{4t^2}\end{aligned}$$

As $t \to \infty$, $\frac{\ln t}{t^2} \to 0$, since a positive power of t grows faster than $\ln t$. Also $\frac{1}{t^2} \to 0$. Therefore

$$\lim_{t\to\infty} \int_e^t \frac{\ln x}{x^3}\,\mathrm{d}x = \frac{3}{4e^2} - 0 - 0 = \frac{3}{4e^2}$$

The limit exists and is finite, so the integral converges, with

$$\int_e^\infty \frac{\ln x}{x^3}\,\mathrm{d}x = \frac{3}{4e^2}$$

7. **(a)** Express $9x^2 - 18x + 25$ in the form $a(x+p)^2 + q$, where a, p and q are integers. **[2]**

(b) Hence use a standard integral from the formula book to find

$$\int \frac{1}{\sqrt{9x^2 - 18x + 25}}\,\mathrm{d}x$$ **[2]**

(c) Determine the exact mean value of the function

$$\mathrm{g}(x) = \frac{1}{\sqrt{9x^2 - 18x + 25}}, \qquad 0 \leq x \leq 2$$

giving your answer in the form $A \ln B$, where A and B are rational constants with B an integer. **[3]**

Solution

(a) Take out the factor of 9 from the x terms and complete the square:

$$\begin{aligned} 9x^2 - 18x + 25 &= 9\big(x^2 - 2x\big) + 25 \\ &= 9\big[(x-1)^2 - 1\big] + 25 \\ &= 9(x-1)^2 + 16 \end{aligned}$$

This is in the form $a(x+p)^2 + q$ with $a = 9$, $p = -1$, $q = 16$.

(b) Using part (a), factor out the 9 inside the square root:

$$\begin{aligned} \int \frac{1}{\sqrt{9x^2 - 18x + 25}}\,\mathrm{d}x &= \int \frac{1}{\sqrt{9(x-1)^2 + 16}}\,\mathrm{d}x \\ &= \frac{1}{3}\int \frac{1}{\sqrt{(x-1)^2 + \left(\frac{4}{3}\right)^2}}\,\mathrm{d}x \end{aligned}$$

The standard result from the formula book, with its constant written as k since a is already in use, is

$$\int \frac{1}{\sqrt{x^2 + k^2}}\,\mathrm{d}x = \operatorname{arsinh}\left(\frac{x}{k}\right) + c$$

Applying this with $k = \frac{4}{3}$, and with x replaced by $x - 1$:

$$\begin{aligned} \int \frac{1}{\sqrt{9x^2 - 18x + 25}}\,\mathrm{d}x &= \frac{1}{3}\operatorname{arsinh}\left(\frac{x-1}{\frac{4}{3}}\right) + c \\ &= \frac{1}{3}\operatorname{arsinh}\left(\frac{3(x-1)}{4}\right) + c \end{aligned}$$

(c) The mean value of $\mathrm{g}(x)$ over $x_1 \leq x \leq x_2$ is

$$\frac{1}{x_2 - x_1}\int_{x_1}^{x_2} \mathrm{g}(x)\,\mathrm{d}x$$

So the mean value of g over $0 \leq x \leq 2$ is

$$\frac{1}{2-0}\int_0^2 \mathrm{g}(x)\,\mathrm{d}x = \frac{1}{2} \times \frac{1}{3}\left[\operatorname{arsinh}\left(\frac{3(x-1)}{4}\right)\right]_0^2$$
$$= \frac{1}{6}\left(\operatorname{arsinh}\left(\frac{3}{4}\right) - \operatorname{arsinh}\left(-\frac{3}{4}\right)\right)$$

The formula book gives

$$\operatorname{arsinh} t = \ln\left(t + \sqrt{t^2+1}\right)$$

Since $\sqrt{\frac{9}{16}+1} = \frac{5}{4}$:

$$\operatorname{arsinh}\left(\frac{3}{4}\right) = \ln\left(\frac{3}{4}+\frac{5}{4}\right) = \ln 2$$
$$\operatorname{arsinh}\left(-\frac{3}{4}\right) = \ln\left(-\frac{3}{4}+\frac{5}{4}\right) = \ln\frac{1}{2} = -\ln 2$$

Therefore the mean value is

$$\frac{1}{6}(\ln 2 - (-\ln 2)) = \frac{1}{6} \times 2\ln 2 = \frac{1}{3}\ln 2$$

This is in the form $A \ln B$ with $A = \frac{1}{3}$, $B = 2$.

8. Let $y = \ln(1+x)$.

(a) Show that the mean value of y with respect to x over the interval $0 \leq x \leq e-1$ is

$$\frac{1}{e-1}$$ **[4]**

(b) Find, in exact form, the mean value of x with respect to y over the interval $0 \leq y \leq 1$. **[3]**

Solution

(a) The mean value of $f(x)$ over $a \leq x \leq b$ is

$$\frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x$$

So the integral $\int_0^{e-1} \ln(1+x)\,\mathrm{d}x$ is needed. Integration by parts states

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Take $u = \ln(1+x)$, $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{1+x}$, $v = x$. Note that $1+x > 0$ on the interval. Then

$$\begin{aligned}\int \ln(1+x)\,\mathrm{d}x &= x\ln(1+x) - \int \frac{x}{1+x}\,\mathrm{d}x \\ &= x\ln(1+x) - \int\left(1 - \frac{1}{1+x}\right)\mathrm{d}x \\ &= x\ln(1+x) - x + \ln(1+x) + c\end{aligned}$$

Evaluating between the limits, using $\ln e = 1$ and $\ln 1 = 0$:

$$\begin{aligned}\int_0^{e-1} \ln(1+x)\,\mathrm{d}x &= \Big[(1+x)\ln(1+x) - x\Big]_0^{e-1} \\ &= \big(e\ln e - (e-1)\big) - \big(1\times\ln 1 - 0\big) \\ &= e - e + 1 \\ &= 1\end{aligned}$$

Hence the mean value is

$$\frac{1}{(e-1)-0}\times 1 = \frac{1}{e-1}$$

as required.

(b) Rearranging $y = \ln(1+x)$:

$$1 + x = e^y \quad\Longrightarrow\quad x = e^y - 1$$

The mean value of x with respect to y over $0 \leq y \leq 1$ is

$$\begin{aligned}\frac{1}{1-0}\int_0^1 (e^y - 1)\mathrm{d}y &= \Big[e^y - y\Big]_0^1 \\ &= (e-1) - (1-0) \\ &= e - 2\end{aligned}$$

9. In an electrical circuit, the current I amps at time t seconds is modelled by

$$I = a\sin nt + b\sin 2nt$$

where a, b and n are positive constants. The RMS value of the current, in amps, is defined to be the square root of the mean value of I^2 over the interval $0 \leq t \leq \frac{2\pi}{n}$.

Show that the RMS value of the current is

$$\sqrt{\frac{a^2+b^2}{2}}$$

amps. **[7]**

Solution

The mean value of $f(t)$ over $p \leq t \leq q$ is

$$\frac{1}{q-p}\int_p^q f(t)\,\mathrm{d}t$$

So the mean value of I^2 over $0 \leq t \leq \frac{2\pi}{n}$ is

$$\frac{n}{2\pi}\int_0^{2\pi/n} I^2\,\mathrm{d}t$$

Expanding I^2:

$$I^2 = a^2\sin^2 nt + 2ab\sin nt\sin 2nt + b^2\sin^2 2nt$$

Using the double angle formula $\cos 2x = 1 - 2\sin^2 x$, we have $\sin^2 x = \frac{1}{2}(1-\cos 2x)$. Hence

$$\begin{aligned}\int_0^{2\pi/n}\sin^2 nt\,\mathrm{d}t &= \int_0^{2\pi/n}\frac{1}{2}(1-\cos 2nt)\mathrm{d}t\\ &= \left[\frac{t}{2}-\frac{\sin 2nt}{4n}\right]_0^{2\pi/n}\\ &= \left(\frac{\pi}{n}-\frac{\sin 4\pi}{4n}\right)-0\\ &= \frac{\pi}{n}\end{aligned}$$

Similarly

$$\begin{aligned}\int_0^{2\pi/n}\sin^2 2nt\,\mathrm{d}t &= \int_0^{2\pi/n}\frac{1}{2}(1-\cos 4nt)\mathrm{d}t\\ &= \left[\frac{t}{2}-\frac{\sin 4nt}{8n}\right]_0^{2\pi/n}\\ &= \left(\frac{\pi}{n}-\frac{\sin 8\pi}{8n}\right)-0\\ &= \frac{\pi}{n}\end{aligned}$$

For the cross term, use the double angle formula $\sin 2nt = 2\sin nt \cos nt$:

$$\begin{aligned}\int_0^{2\pi/n} \sin nt \sin 2nt \,\mathrm{d}t &= \int_0^{2\pi/n} 2\sin^2 nt \cos nt \,\mathrm{d}t \\ &= \left[\frac{2\sin^3 nt}{3n}\right]_0^{2\pi/n} \\ &= \frac{2\sin^3 2\pi}{3n} - 0 \\ &= 0\end{aligned}$$

Combining these results:

$$\begin{aligned}\int_0^{2\pi/n} I^2 \,\mathrm{d}t &= a^2 \times \frac{\pi}{n} + 2ab \times 0 + b^2 \times \frac{\pi}{n} \\ &= \frac{\left(a^2 + b^2\right)\pi}{n}\end{aligned}$$

So the mean value of I^2 is

$$\frac{n}{2\pi} \times \frac{\left(a^2 + b^2\right)\pi}{n} = \frac{a^2 + b^2}{2}$$

Taking the square root, the RMS value of the current is

$$\sqrt{\frac{a^2 + b^2}{2}}$$

amps, as required.

10. Prove that

$$\int_1^\infty \frac{x+3}{x^2(x+1)}\,\mathrm{d}x = 3 - \ln 4$$

Show the limiting process clearly. **[8]**

Solution

The integral is improper, so it is defined as

$$\int_1^\infty \frac{x+3}{x^2(x+1)}\,\mathrm{d}x = \lim_{t\to\infty}\int_1^t \frac{x+3}{x^2(x+1)}\,\mathrm{d}x$$

First write the integrand in partial fractions:

$$\frac{x+3}{x^2(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} \quad\Longrightarrow\quad x+3 = Ax(x+1) + B(x+1) + Cx^2$$

Setting $x = 0$ gives $B = 3$. Setting $x = -1$ gives $2 = C$. Equating coefficients of x^2 gives $A + C = 0$, so $A = -2$. Hence

$$\frac{x+3}{x^2(x+1)} = -\frac{2}{x} + \frac{3}{x^2} + \frac{2}{x+1}$$

Since $x \geq 1$, both x and $x+1$ are positive, so no modulus signs are needed in the logarithms. For $t > 1$:

$$\begin{aligned}\int_1^t \left(-\frac{2}{x} + \frac{3}{x^2} + \frac{2}{x+1}\right)\mathrm{d}x &= \left[-2\ln x - \frac{3}{x} + 2\ln(x+1)\right]_1^t \\ &= \left[2\ln\left(\frac{x+1}{x}\right) - \frac{3}{x}\right]_1^t \\ &= \left(2\ln\left(1+\frac{1}{t}\right) - \frac{3}{t}\right) - (2\ln 2 - 3)\end{aligned}$$

The logarithms are combined before taking the limit, since $\ln t$ and $\ln(t+1)$ separately both tend to infinity. As $t \to \infty$, $\frac{1}{t} \to 0$, so $1 + \frac{1}{t} \to 1$ and $\ln\left(1+\frac{1}{t}\right) \to \ln 1 = 0$. Also $\frac{3}{t} \to 0$. Therefore

$$\begin{aligned}\int_1^\infty \frac{x+3}{x^2(x+1)}\,\mathrm{d}x &= \lim_{t\to\infty}\left(2\ln\left(1+\frac{1}{t}\right) - \frac{3}{t} - 2\ln 2 + 3\right) \\ &= 0 - 0 - 2\ln 2 + 3 \\ &= 3 - \ln 4\end{aligned}$$

as required.

11. **(a)** Explain why

$$\int_0^\infty \frac{10}{(x+2)(3x+1)}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) Show that

$$\int_0^\infty \frac{10}{(x+2)(3x+1)}\,\mathrm{d}x = a\ln b$$

where a is an integer and b is a rational number, both to be determined. You must show clearly how you deal with the infinite limit. **[6]**

Solution

(a) The upper limit of integration is infinite, so the interval of integration is unbounded and the integral is improper.

(b) First express the integrand in partial fractions:

$$\frac{10}{(x+2)(3x+1)} = \frac{A}{x+2} + \frac{B}{3x+1}$$

so

$$10 = A(3x+1) + B(x+2)$$

Setting $x = -2$ gives $10 = -5A$, so $A = -2$. Setting $x = -\frac{1}{3}$ gives $10 = \frac{5}{3}B$, so $B = 6$.

For $x \geq 0$, both $x+2$ and $3x+1$ are positive, so

$$\begin{aligned}\int \frac{10}{(x+2)(3x+1)}\,\mathrm{d}x &= \int\left(-\frac{2}{x+2} + \frac{6}{3x+1}\right)\mathrm{d}x \\ &= -2\ln(x+2) + 2\ln(3x+1) + c \\ &= 2\ln\left(\frac{3x+1}{x+2}\right) + c\end{aligned}$$

Replace the infinite limit by t, where $t > 0$:

$$\begin{aligned}\int_0^t \frac{10}{(x+2)(3x+1)}\,\mathrm{d}x &= \left[2\ln\left(\frac{3x+1}{x+2}\right)\right]_0^t \\ &= 2\ln\left(\frac{3t+1}{t+2}\right) - 2\ln\left(\frac{1}{2}\right) \\ &= 2\ln\left(\frac{3t+1}{t+2}\right) + 2\ln 2\end{aligned}$$

Dividing the numerator and denominator by t:

$$\frac{3t+1}{t+2} = \frac{3+\frac{1}{t}}{1+\frac{2}{t}} \to \frac{3+0}{1+0} = 3 \quad \text{as } t \to \infty$$

so $2\ln\left(\frac{3t+1}{t+2}\right) \to 2\ln 3$. Therefore

$$\begin{aligned}\int_0^\infty \frac{10}{(x+2)(3x+1)}\,\mathrm{d}x &= \lim_{t\to\infty}\left(2\ln\left(\frac{3t+1}{t+2}\right)+2\ln 2\right)\\ &= 2\ln 3 + 2\ln 2\\ &= 2\ln 6\end{aligned}$$

This is in the form $a\ln b$ with $a = 2$, $b = 6$.

12. **(a)** Explain why

$$\int_0^3 \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) Use an appropriate limit argument to show that this integral converges, and find its exact value. **[5]**

Solution

(a) When $x = 1$, $x^2 - 1 = 0$, so the denominator $\sqrt[3]{x^2-1}$ is zero and the integrand is undefined (it is unbounded near $x = 1$). Since $x = 1$ lies inside the interval of integration $[0, 3]$, the integral is improper.

(b) First find an antiderivative. Write the integrand as $x\left(x^2-1\right)^{-\frac{1}{3}}$, taking real cube roots, which exist for negative numbers too: for example $\sqrt[3]{-1} = -1$. By the chain rule, for $x \neq 1$:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt[3]{x^2-1}\right)^2 = \frac{\mathrm{d}}{\mathrm{d}x}\left(x^2-1\right)^{\frac{2}{3}} = \frac{2}{3}\left(x^2-1\right)^{-\frac{1}{3}} \times 2x = \frac{4}{3}\frac{x}{\sqrt[3]{x^2-1}}$$

Hence

$$\int \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x = \frac{3}{4}\left(\sqrt[3]{x^2-1}\right)^2 + c$$

Split the integral at the point where the integrand is undefined:

$$\int_0^3 \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x = \int_0^1 \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x + \int_1^3 \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x$$

The original integral converges if and only if both of these converge.

For the first integral, with $0 < t < 1$:

$$\begin{aligned}
\int_0^1 \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x &= \lim_{t\to 1^-}\left[\frac{3}{4}\left(\sqrt[3]{x^2-1}\right)^2\right]_0^t \\
&= \lim_{t\to 1^-}\left(\frac{3}{4}\left(\sqrt[3]{t^2-1}\right)^2 - \frac{3}{4}\left(\sqrt[3]{-1}\right)^2\right) \\
&= 0 - \frac{3}{4}\times(-1)^2 \\
&= -\frac{3}{4}
\end{aligned}$$

since $t^2 - 1 \to 0$ as $t \to 1^-$. This is finite, so the first integral converges.

For the second integral, with $1 < s < 3$:

$$\begin{aligned}
\int_1^3 \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x &= \lim_{s\to 1^+}\left[\frac{3}{4}\left(\sqrt[3]{x^2-1}\right)^2\right]_s^3 \\
&= \lim_{s\to 1^+}\left(\frac{3}{4}\left(\sqrt[3]{8}\right)^2 - \frac{3}{4}\left(\sqrt[3]{s^2-1}\right)^2\right) \\
&= \frac{3}{4}\times 4 - 0 \\
&= 3
\end{aligned}$$

This is also finite, so the second integral converges.

Both limits exist and are finite, so the integral converges, and

$$\int_0^3 \frac{x}{\sqrt[3]{x^2-1}}\,\mathrm{d}x = -\frac{3}{4} + 3 = \frac{9}{4}$$

13. **(a)** Explain why

$$\int_0^e \frac{(\ln x)^2}{\sqrt{x}}\,\mathrm{d}x$$

is an improper integral. **[1]**

(b) By using integration by parts twice, show that

$$\int x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x = 2\sqrt{x}\left((\ln x)^2 - 4\ln x + 8\right) + c \qquad \textbf{[4]}$$

(c) You may assume that, for any constant $k > 0$ and any positive integer n, $x^k(\ln x)^n \to 0$ as $x \to 0^+$.

Show that

$$\int_0^e \frac{(\ln x)^2}{\sqrt{x}}\,\mathrm{d}x$$

exists and find its exact value. **[4]**

Solution

(a) The integrand is not defined at the lower limit $x = 0$. As $x \to 0^+$, $(\ln x)^2 \to \infty$ and $\sqrt{x} \to 0$, so $\frac{(\ln x)^2}{\sqrt{x}} \to \infty$. The integrand is unbounded on the interval of integration, so the integral is improper.

(b) Integration by parts states

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

First take $u = (\ln x)^2$, $\frac{\mathrm{d}v}{\mathrm{d}x} = x^{-\frac{1}{2}}$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2\ln x}{x}$, $v = 2x^{\frac{1}{2}}$. Then

$$\begin{aligned}\int x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x &= 2x^{\frac{1}{2}}(\ln x)^2 - \int 2x^{\frac{1}{2}} \times \frac{2\ln x}{x}\,\mathrm{d}x \\ &= 2\sqrt{x}(\ln x)^2 - 4\int x^{-\frac{1}{2}}\ln x\,\mathrm{d}x\end{aligned}$$

For the remaining integral take $u = \ln x$, $\frac{\mathrm{d}v}{\mathrm{d}x} = x^{-\frac{1}{2}}$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$, $v = 2x^{\frac{1}{2}}$. Then

$$\begin{aligned}\int x^{-\frac{1}{2}}\ln x\,\mathrm{d}x &= 2x^{\frac{1}{2}}\ln x - \int 2x^{\frac{1}{2}} \times \frac{1}{x}\,\mathrm{d}x \\ &= 2\sqrt{x}\ln x - 2\int x^{-\frac{1}{2}}\,\mathrm{d}x \\ &= 2\sqrt{x}\ln x - 4\sqrt{x}\end{aligned}$$

Combining the two results:

$$\begin{aligned}\int x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x &= 2\sqrt{x}(\ln x)^2 - 4\left(2\sqrt{x}\ln x - 4\sqrt{x}\right) + c \\ &= 2\sqrt{x}(\ln x)^2 - 8\sqrt{x}\ln x + 16\sqrt{x} + c \\ &= 2\sqrt{x}\left((\ln x)^2 - 4\ln x + 8\right) + c\end{aligned}$$

as required.

(c) Since the integrand is unbounded at $x = 0$, replace the lower limit by t, where $0 < t < e$, and let $t \to 0^+$:

$$\int_0^e \frac{(\ln x)^2}{\sqrt{x}}\,\mathrm{d}x = \lim_{t\to 0^+} \int_t^e x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x$$

Using part (b):

$$\begin{aligned}\int_t^e x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x &= \left[2\sqrt{x}\big((\ln x)^2 - 4\ln x + 8\big)\right]_t^e \\ &= 2\sqrt{e}\,(1 - 4 + 8) - 2\sqrt{t}\big((\ln t)^2 - 4\ln t + 8\big) \\ &= 10\sqrt{e} - 2t^{\frac{1}{2}}(\ln t)^2 + 8t^{\frac{1}{2}}\ln t - 16t^{\frac{1}{2}}\end{aligned}$$

As $t \to 0^+$, using the given result with $k = \frac{1}{2}$ and $n = 2$, $t^{\frac{1}{2}}(\ln t)^2 \to 0$; with $k = \frac{1}{2}$ and $n = 1$, $t^{\frac{1}{2}}\ln t \to 0$; and $16t^{\frac{1}{2}} \to 0$. Hence

$$\lim_{t\to 0^+} \int_t^e x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x = 10\sqrt{e} - 0 + 0 - 0 = 10\sqrt{e}$$

The limit is finite, so the integral exists, and

$$\int_0^e \frac{(\ln x)^2}{\sqrt{x}}\,\mathrm{d}x = 10\sqrt{e}$$

14. **(a)** Find

$$\int_1^a \frac{(\ln x)^2}{x^2}\,\mathrm{d}x$$

where $a > 1$, giving your answer in terms of a. **[6]**

(b) Write down the value of

$$\lim_{a\to\infty} \frac{(\ln a)^k}{a}$$

where k is a positive constant. **[1]**

(c) Hence find

$$\int_1^\infty \frac{(\ln x)^2}{x^2}\,\mathrm{d}x$$ **[2]**

Solution

(a) Integration by parts states

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Take $u = (\ln x)^2$, $\frac{\mathrm{d}v}{\mathrm{d}x} = x^{-2}$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2\ln x}{x}$, $v = -\frac{1}{x}$. Then

$$\begin{aligned}\int \frac{(\ln x)^2}{x^2}\,\mathrm{d}x &= -\frac{(\ln x)^2}{x} - \int\left(-\frac{1}{x}\right)\frac{2\ln x}{x}\,\mathrm{d}x\\ &= -\frac{(\ln x)^2}{x} + 2\int \frac{\ln x}{x^2}\,\mathrm{d}x\end{aligned}$$

For the remaining integral, apply integration by parts again with $u = \ln x$, $\frac{\mathrm{d}v}{\mathrm{d}x} = x^{-2}$, so $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$, $v = -\frac{1}{x}$:

$$\begin{aligned}\int \frac{\ln x}{x^2}\,\mathrm{d}x &= -\frac{\ln x}{x} - \int\left(-\frac{1}{x}\right)\frac{1}{x}\,\mathrm{d}x\\ &= -\frac{\ln x}{x} + \int x^{-2}\,\mathrm{d}x\\ &= -\frac{\ln x}{x} - \frac{1}{x} + c\end{aligned}$$

Combining these:

$$\int \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = -\frac{(\ln x)^2}{x} - \frac{2\ln x}{x} - \frac{2}{x} + c$$

Evaluating between the limits, using $\ln 1 = 0$:

$$\begin{aligned}\int_1^a \frac{(\ln x)^2}{x^2}\,\mathrm{d}x &= \left[-\frac{(\ln x)^2}{x} - \frac{2\ln x}{x} - \frac{2}{x}\right]_1^a\\ &= \left(-\frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}\right) - (0 - 0 - 2)\\ &= 2 - \frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}\end{aligned}$$

(b) Any power of $\ln a$ grows more slowly than a, so

$$\lim_{a\to\infty} \frac{(\ln a)^k}{a} = 0$$

(c) The improper integral is defined as a limit, so using part (a):

$$\int_1^\infty \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = \lim_{a\to\infty}\left(2 - \frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}\right)$$

By part (b) with $k = 2$ and with $k = 1$, $\frac{(\ln a)^2}{a} \to 0$ and $\frac{\ln a}{a} \to 0$ as $a \to \infty$; also $\frac{2}{a} \to 0$. Therefore

$$\int_1^\infty \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = 2 - 0 - 0 - 0 = 2$$

15. Show that

$$\int_0^\infty \frac{5x-3}{(x^2+3)(x+1)}\,\mathrm{d}x = p\sqrt{3}\,\pi + \ln q$$

where p and q are rational numbers to be found. **[8]**

Solution

First express the integrand in partial fractions:

$$\frac{5x-3}{(x^2+3)(x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+3}$$

Multiplying through by $(x^2+3)(x+1)$ gives

$$5x-3 = A(x^2+3) + (Bx+C)(x+1)$$

Setting $x=-1$:

$$-8 = 4A \quad\Longrightarrow\quad A = -2$$

Equating coefficients of x^2 and the constant terms:

$$x^2: \quad 0 = A + B \quad\Longrightarrow\quad B = 2$$

$$\text{constant}: \quad -3 = 3A + C \quad\Longrightarrow\quad C = 3$$

So

$$\frac{5x-3}{(x^2+3)(x+1)} = -\frac{2}{x+1} + \frac{2x}{x^2+3} + \frac{3}{x^2+3}$$

For the last term, the standard result is

$$\int \frac{1}{x^2+a^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + c$$

with $a=\sqrt{3}$. Since $x+1>0$ for $x\geq 0$, the logarithm needs no modulus, so

$$\begin{aligned}\int \frac{5x-3}{(x^2+3)(x+1)}\,\mathrm{d}x &= -2\ln(x+1) + \ln(x^2+3) + \frac{3}{\sqrt{3}}\arctan\left(\frac{x}{\sqrt{3}}\right) + c \\ &= \ln\left(\frac{x^2+3}{(x+1)^2}\right) + \sqrt{3}\arctan\left(\frac{x}{\sqrt{3}}\right) + c\end{aligned}$$

The integral is improper, so write it as a limit:

$$\begin{aligned}\int_0^\infty \frac{5x-3}{(x^2+3)(x+1)}\,\mathrm{d}x &= \lim_{t\to\infty}\left[\ln\left(\frac{x^2+3}{(x+1)^2}\right) + \sqrt{3}\arctan\left(\frac{x}{\sqrt{3}}\right)\right]_0^t \\ &= \lim_{t\to\infty}\left[\ln\left(\frac{t^2+3}{(t+1)^2}\right) + \sqrt{3}\arctan\left(\frac{t}{\sqrt{3}}\right)\right] - \left(\ln 3 + \sqrt{3}\arctan 0\right)\end{aligned}$$

Dividing the numerator and denominator by t^2:

$$\frac{t^2+3}{(t+1)^2} = \frac{1+\frac{3}{t^2}}{\left(1+\frac{1}{t}\right)^2} \to 1 \quad \text{as } t\to\infty$$

so $\ln\left(\frac{t^2+3}{(t+1)^2}\right) \to \ln 1 = 0$. Also $\arctan\left(\frac{t}{\sqrt{3}}\right) \to \frac{\pi}{2}$, and $\arctan 0 = 0$. Therefore

$$\begin{aligned}\int_0^\infty \frac{5x-3}{(x^2+3)(x+1)}\,\mathrm{d}x &= 0 + \frac{\sqrt{3}\,\pi}{2} - \ln 3 \\ &= \frac{1}{2}\sqrt{3}\,\pi + \ln\frac{1}{3}\end{aligned}$$

This is in the form $p\sqrt{3}\,\pi + \ln q$ with $p = \frac{1}{2}$ and $q = \frac{1}{3}$.

16. **(a)** Explain why

$$\int_0^1 (\ln y)^2 \, dy$$

is an improper integral. **[1]**

(b) (i) Show that the substitution $y = e^{-x}$ transforms

$$\int_0^1 (\ln y)^2 \, dy \quad \text{into} \quad \int_0^\infty x^2 e^{-x} \, dx$$

explaining clearly how the limits of integration are obtained. **[3]**

(ii) Evaluate

$$\int_0^\infty x^2 e^{-x} \, dx$$

showing the limiting process used. **[6]**

(iii) Hence write down the value of

$$\int_0^1 (\ln y)^2 \, dy$$ **[1]**

Solution

(a) As $y \to 0^+$, $\ln y \to -\infty$, so $(\ln y)^2 \to \infty$. The integrand is therefore unbounded (and undefined) at the lower limit $y = 0$, so the integral is improper.

(b) (i) With $y = e^{-x}$, we have $\ln y = -x$, so $(\ln y)^2 = x^2$. Also

$$\frac{dy}{dx} = -e^{-x} \quad \Longrightarrow \quad dy = -e^{-x} \, dx$$

For the limits: when $y = 1$, $e^{-x} = 1$, so $x = 0$. The lower limit is improper, so write it as $y = s$ and let $s \to 0^+$. Then $x = -\ln s$, and as $s \to 0^+$, $-\ln s \to \infty$. So the lower limit $y \to 0^+$ corresponds to $x \to \infty$.

Hence

$$\begin{aligned} \int_0^1 (\ln y)^2 \, dy &= \int_\infty^0 x^2 \left(-e^{-x}\right) dx \\ &= \int_0^\infty x^2 e^{-x} \, dx \end{aligned}$$

as required, where reversing the limits removes the minus sign.

(ii) The integral is defined as

$$\int_0^\infty x^2 e^{-x} \, dx = \lim_{t \to \infty} \int_0^t x^2 e^{-x} \, dx$$

First find the indefinite integral using integration by parts:

$$\int u \frac{dv}{dx} \, dx = uv - \int v \frac{du}{dx} \, dx$$

Take $u = x^2$, $\frac{dv}{dx} = e^{-x}$, so $\frac{du}{dx} = 2x$, $v = -e^{-x}$:

$$\int x^2 e^{-x}\,dx = -x^2 e^{-x} + \int 2xe^{-x}\,dx$$

For the remaining integral take $u = 2x$, $\frac{dv}{dx} = e^{-x}$, so $\frac{du}{dx} = 2$, $v = -e^{-x}$:

$$\begin{aligned}\int 2xe^{-x}\,dx &= -2xe^{-x} + \int 2e^{-x}\,dx \\ &= -2xe^{-x} - 2e^{-x} + c\end{aligned}$$

Therefore

$$\int x^2 e^{-x}\,dx = -e^{-x}\left(x^2 + 2x + 2\right) + c$$

So

$$\begin{aligned}\int_0^t x^2 e^{-x}\,dx &= \left[-e^{-x}\left(x^2 + 2x + 2\right)\right]_0^t \\ &= -e^{-t}\left(t^2 + 2t + 2\right) - \left(-e^0 \times 2\right) \\ &= 2 - t^2 e^{-t} - 2te^{-t} - 2e^{-t}\end{aligned}$$

As $t \to \infty$, the exponential dominates any power of t, so $t^2 e^{-t} \to 0$ and $te^{-t} \to 0$; also $e^{-t} \to 0$. Hence

$$\int_0^\infty x^2 e^{-x}\,dx = \lim_{t\to\infty}\left(2 - t^2 e^{-t} - 2te^{-t} - 2e^{-t}\right) = 2$$

(iii) By part (b)(i) the two integrals are equal, so by part (b)(ii)

$$\int_0^1 (\ln y)^2\,dy = 2$$